

Are we missing the ‘care’ in Healthcare AI?

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While there are many ethical concerns with healthcare AI, most critiques rarely mention the role of emotions in clinical care and its implications for AI development, which I argue is especially important to consider. I outline the role of emotional intelligence in building patient-doctor relationships, which helps improve clinical decision-making and patient outcomes. Affects are crucial to the doctor-patient relationship and thus crucial to consider when building healthcare AI. As such, I suggest two potential ways forward: 1) building affective health AI or, alternatively, 2) refocusing efforts on building AI tools that enhance the doctor-patient relationship.

Additional Key Words and Phrases: affective computing, doctor-patient relationship, clinical decision-making, emotional intelligence

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1 Introduction

‘Healthcare AI’ refers to the subset of artificial intelligence (AI) technologies that aim to improve the medical field across various domains including diagnostics, personalized medicine, and patient care. Recently, there has been a growing interest in AI for clinical decision-making, with the potential to transform clinical care through increased efficiency, accuracy, and personalization [10]. This includes IBM’s Watson Oncology [7] which predicts the most effective cancer treatments for patients, and medical chatbots [5, 16] that answer patients’ questions and help diagnose rare diseases in hospitals. Despite the potential benefits, there are many ethical concerns. Current healthcare AI models are unreliable, such as Microsoft’s BioGPT which provides pseudo-scientific or inaccurate answers about medicine and biology [22]; other models further increase health disparities due to algorithmic bias [19]. Similarly, large language models (LLMs) have been shown to propagate race-based medicine [20], and their limitations in clinical settings have not been thoroughly researched despite their increased use [24].

Now, even if we were able to solve these issues of diagnostic AI systems, and had ‘better than human’ accuracy, would this result in a positive transformation to the healthcare system? I argue that this is not the case because providing patients with diagnoses is not the sole purpose of a doctor’s appointment. Recent shifts to patient-centred care instead emphasize *relational moral competency* including empathy, compassion and trust, which is dependent on the doctor’s competency as well as their interpersonal and emotional skills [21]. Thus, the caregiver’s priority is to understand the patient’s perspective and guide them to make the right decision based on their needs, desires, and ideals, which are deeply influenced by their affects [10].

This position paper explores the importance of affects in doctor-patient relationships, and how it should impact the development of healthcare AI. In Section 2, I highlight the inseparable nature of cognitive and affective processes in holistic patient care. In Section 3, I explain why a purely rational AI system that influences clinical decision-making would result in poorer health outcomes, regardless of whether it replaced physicians or was used as a tool. Finally, in Section 4, I propose two ways forward to mitigate these concerns: (a) develop affective AI systems for clinical decision-making or, alternatively, (b) shift focus to AI tools for non-clinical tasks in healthcare so that physicians can focus on providing holistic patient care.

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2 The Role of Affect and Human Connection in Clinical Care

Historically, medicine was believed to be a purely objective field, based on technical expertise where physicians had full decision-making power [10, 12, 14]. This paternalist mode of care disregarded the doctor-patient relationship, the patient's values and their autonomy [14]. However, recent shifts to patient-centered care highlighted the importance of shared clinical decision-making. Now, the *care concept* includes a holistic approach rather than just the pathological aspects [1]. This requires healthcare practitioners to go beyond symptoms and do the emotional work to appraise patients' affects (e.g., anxiety, sadness, hope, anger) [6].

Inseparable nature of affects and clinical care. As Paul Burcher states, "the clinical encounter, or the relation between physician and patient is perhaps unique in that the exchange and processing of emotionally laden information must often occur rapidly and between two people who may be near strangers to one another" [15]. Clinical judgments cannot be reduced to algorithmic decisions, as they require complex integration of medical knowledge with emotional understanding [9]. This integration is essential since clinical decisions are a mix of emotion-based and cognitive-based processing [6, 12]. If a clinician were to suppress their emotions, it may result in them missing or distorting evidence needed to make a sound clinical decision [15]. Social and cognitive psychology research shows that emotions play an important role largely because medical care goes beyond a diagnosis [12]. Take, for instance, Verghese's account of *bedside as a ritual*: "the physical exam... is much more than a means of diagnosing disease; rather, it is a transformative ritual that signifies the physicians' care for their patients and their commitment to be with them through their illness" [9]. Similarly, many physicians and hospital workers emphasized the physician's role in engaging with patients through consultations and consolations, especially for those who needed space to "vent" [25].

Emotional intelligence (EI) as a core element of holistic medical care. EI is a crucial skill for healthcare providers since they have to carefully manage patient's emotions [4]. In fact, high EI has been shown to improve medical students' interpersonal and communication skills, one of the six core educational competencies in medicine [15]. Similarly, patients prefer to engage with physicians who have the right interpersonal and emotional skills [10]. An emotionally intelligent physician can connect with the patient in order to communicate effectively. Empathy is a core component of EI, and is essential to providing *deep care* which goes "beyond the mere fulfillment of medical tasks" [8]. Empathic clinicians gather vital patient information more effectively, which influences the quality of care delivered and overall medical outcomes. Empathy has been shown to decrease patient anxiety while its absence can lead to biased decision-making [7]. Similarly, compassionate caregiving involves both sensitivity to one's suffering (i.e., empathy) and the commitment to try to prevent or alleviate said suffering [1, 10, 18]. Physicians' empathy helps them understand and share patients' feelings which, in turn, increases the patients' involvement in the decision-making process. Compassion is instigated by this empathetic engagement with the patient, leading to the desire to help [15].

The therapeutic clinical relationship between doctor and patient. This relationship is defined by the doctor's intent of improving care, where "one must be able to interact and treat the patient and not just the disease" [11, 25]. Take, for example, a patient speaking about the friendship they have built with their doctor: "When I come to her, it's like talking to a friend – we chat a little, laugh, and then I start talking about my [health] problems" [25]. Inversely, lacking therapeutic relationships can adversely impact the patient, as is the case with Mrs. Jones, a care facility resident whose clinical nurse recently left: "I cried and cried. People didn't seem to understand how important she had been to me. I've no one to talk to now. They just rush in and out and get things over with as quickly as possible" [7]. The emotional dimension of an interaction helps form robust clinical relationships, and dismissing that aspect can lead to inferior patient outcomes [15]. This relationship is dependent on trust, which is built through several factors: (a)

the patient must have faith in the doctor’s competence, (b) the doctor must know and understand patient’s values and beliefs, and (c) the doctor must trust that the patient is, to the best of their abilities, accurately reporting symptoms [11, 15]. The clinician must also be able to empathize with the patient’s experiences, concerns, and goals [11]. This relies on the physician’s ability to both appraise and express emotions to a patient. The therapeutic relationship also improves patient satisfaction, and has been shown to improve recovery, treatment adherence, and follow-up visits [11].

3 Potential Impacts of AI on the Emotional Aspects of Care

Now, let us consider how purely rational ‘diagnostic AI’ (i.e., without affective reasoning) may impact healthcare. This includes screening patients for risk/severity, interpreting symptoms for diagnosis, and developing treatment plans, which accounts for most applications in medicine today [10]. One optimistic scenario is that diagnostic AI could benefit patient care, if integrated carefully enough [1, 2]. For example, Eric Schmidt, former chair of Alphabet, states that “if the promise of AI is fulfilled, we should be able to concentrate more fully on engaging with our patients” [3]. However, this hope that diagnostic AI provide technical support while physicians provide the ‘human touch’ is infeasible.

Diagnostic AI replacing physicians. If AI could complete most tasks performed by physicians, the incentives for involving humans would drastically decrease. Considering the supposed cost-effectiveness of AI solutions while hospitals are currently under pressure to cut costs [7], it would be economically disadvantageous to integrate diagnostic AI technologies while also paying physicians. Doctors are already incentivized to increase the number of patients they see to cover expenses [11]. At the very least, hospitals will no longer have doctors with the same skills and training. In the worst-case, diagnostic AI would render physicians obsolete because hospitals would no longer see the need for them. Ultimately, it seems that “techno-commercial motives are discordant with professional-relational foundation for care” [18]. If this were to happen, affects would no longer be considered in the clinical decision-making process and the field of healthcare would regress back to the paternalistic mode of care, where “the computer knows best” [14]. Patients require counselling and discussion and, without it, bad decision-making would occur. For instance, at the same stage of a terminal disease, one patient might choose palliative care whereas another would opt for further therapy. If the goal of diagnostic AI was to maximize lifespan, it would result in a negative outcome for the first patient. There is no ‘one size fits all’ solution in clinical settings since “decisions are not simply a matter of survival-based logic” [14] – decisions should be made in the context of the patient’s lived experiences. Diagnostic AI would also lead to the erosion of empathy and the doctor-patient relationship. Many stakeholders are concerned about losing the ‘human touch’ because AI cannot provide, for example, warm words and communication which are often crucial to patients [18, 25]. Without considering patients’ emotions, healthcare would reduce to the mere completion of medical tasks.

Diagnostic AI as a tool for physicians. Rather than replacing physicians, suppose that diagnostic AI was used as a tool. In this scenario, its integration would further increase doctors’ workload. There is already a *care gap* in the field due to underpaid and overworked physicians, where they spend more time completing administrative tasks rather than with their patients [10, 18, 25]. If current medical technologies hinder the patient-doctor relationship, how can we expect otherwise with future technologies? Many physicians are concerned that AI would create more work for them, where they would have to learn to interact with and evaluate AI systems [2, 21]. It is paradoxical - if the promise of increased efficiency results in physicians taking up additional tasks, where are they supposed to conjure up the time to engage with their patients? There is also risk of over-reliance. The AI’s recommendations might influence physicians’ decisions more than they realize, leading to a more *transactional approach* to healthcare, where empathy and understanding are diminished and doctors turn into “messengers of AI’s outputs” [1, 14]. This level of AI involvement would prioritize task completion over the patient-doctor relationship, resulting in disengaged patients who feel dehumanized, “as if

they are being reduced to mere statistical figures” [1]. This could also raise questions about who is truly guiding the decision-making process, potentially leading to doubts about the physician’s competency, a fundamental characteristic of a trusting relationship [10]. Lastly, the emotional tasks of patient care are still inseparable from the cognitive. *Deep care* is the combination of performing medical tasks while caring about the patient [7]. A physician could provide the labour of care without the caring attitude but this would not be *good* care since “the open responsiveness to another that is so essential to understanding what another requires is not possible” [7]. Here, the clinician did not acquire the skills to fully understand or address the patient’s needs. Thus, the labour and attitude of care must be performed by the same carer (i.e., human or AI).

4 Implications for AI Development in Healthcare

As outlined above, affects are an integral part of healthcare, and patient care cannot simply be separated into the rational AI and caring human. Thus, there is a need for alternative ways forward in the field of healthcare AI. In this section, I propose two potential solutions.

Developing affective healthcare AI. One solution is to develop emotionally intelligent AI. In this scenario, AI would both appraise and express emotions, essentially replacing physicians in patient care. This would preserve the ideal of patient-centered care, allowing patients’ differing values, desires and priorities to be considered by AI [10, 21]. Here, patients would learn about their diagnoses from a compassionate AI and collaborate with it to develop a treatment plan. This seems plausible based on pre-existing work on artificial empathy, where technologies are designed to sense and display empathy to their users [18]. OpenAI’s GPT-4 also seems to be capable of identifying emotions and describing emotional management techniques [23]. Artificial compassion would further require researchers to intentionally design “adaptive responsiveness into technologies” [18]. However, it seems unlikely that truly emotionally intelligent AI will exist in the near future. Currently, applications of affective computing in medicine focus on a narrow set of specialties - specifically, mental health disorders, autism, and pain level recognition [13]. Furthermore, these systems tend to rely on simplistic 3- or 6-class models and are unable to provide accurate real-time emotional analysis [13]. Some scholars go further to state that AI is incapable of understanding empathy, and thus should never replace medical professionals [7, 17]. Hirmiz states that patients simply believing that AI cares for them would not be satisfactory [7], which is supported by a study demonstrating that most patients prefer to follow the advice of a doctor rather than personalized AI [21]. Most stakeholders prefer AI systems that support, rather than replace, the human aspects of patient care.

Developing AI for non-clinical tasks. This leads to our second proposed solution, abandoning the development of AI for clinical decision-making altogether. Considering that the viability and existence of emotionally intelligent healthcare AI is still up for debate, it might make more sense to refocus efforts on developing AI for non-clinical, administrative tasks so that doctors can focus on building a therapeutic relationship with their patients. Take, for example, this physician’s account: “Honestly, all that typing, printing, and confirming of test results and such, I waste a lot of time on it, and the actual examination of the patient and talking to them, I believe, should be retained, but in a way that avoids the need for me to look at the screen constantly. Personally, when I gather data, sometimes I forget that, while I’m typing and looking at the screen, I’m not really looking at the patient themselves, and I end up missing information I could gather just by observing them” [25]. Administrative tasks such as manual data collection during consultations are one of the most time-consuming and have been shown to erode the doctor-patient relationship [18]. In fact, a study from 2018 stated that UK’s national health system could save “up to 10% of its running costs by outsourcing repetitive and administrative tasks to artificial intelligence technologies” [10]. Many stakeholders also hope that AI integration will address other non-clinical tasks such as decreasing waiting times and overcrowding, and

improving access to healthcare services [25]. Such applications of AI may be able to address the care gap prevalent in healthcare today [18]. The promise of greater efficiency with the integration of AI in healthcare is possible, but only if the right tasks are automated. Therefore, researchers should focus on creating tools that help healthcare workers improve relationships with their patients, rather than drifting them further apart.

5 Conclusion

In this paper, I hope to have outlined that affects are crucial to the clinical decision-making process and thus crucial to consider when building healthcare AI systems. Diagnosis is not the be-all end-all of healthcare - empathy, compassion, and trust are an integral part of the patient-doctor relationship which helps improve clinical outcomes. The development of a purely rational diagnostic AI would lead to the erosion of empathy and trust in clinical settings, resulting in a more transactional approach to healthcare that disregards the patient’s emotional needs, values and desires. Therefore, I propose two potential solutions that account for the role of affects in clinical care. First, researchers could incorporate affective computing techniques to develop emotionally intelligent AI systems for clinical decision-making. However, the existence of truly emotionally intelligent AI seems unlikely in the near future and this may not be desirable for patients, or society, after all. Thus, alternatively, researchers should increase focus on developing AI for non-clinical, administrative tasks which would allow physicians to focus on fostering therapeutic relationships with patients.

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